



वडोदा RACING
LIVE IT. LOVE IT. RACE IT.

ENGINEERING
ELECTRIC
PERFORMANCE.



**FEBRUARY
2026
EDITION**

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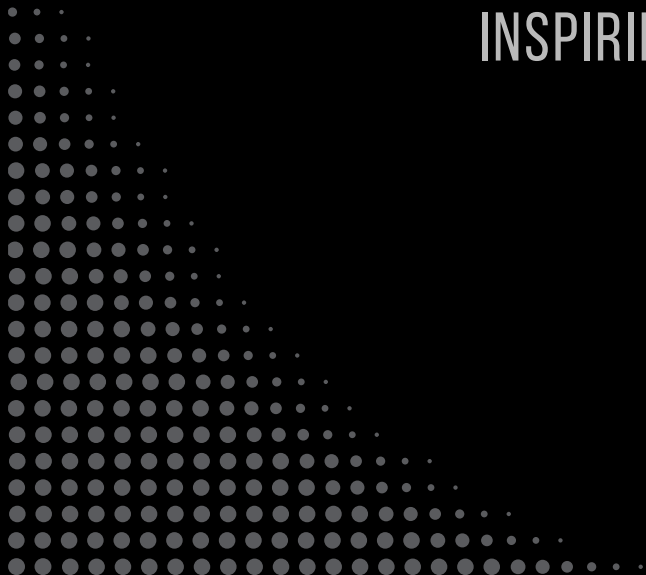


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**VISION
AND
MISSION**

GUIDING PURPOSE,
INSPIRING FUTURE.



> MISSION AND VISION <

VISION

- To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.
- To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

- Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.
- Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.



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**VOICES FROM
THE
INDUSTRY**

INDUSTRY VISITS
THE WORKSHOP





VOICES FROM THE INDUSTRY

The Vision Behind Ashwa Racing – A Seminar with Founder Mr. Arunraj Subbaraj

Mr. Arunraj Subbaraj, established Ashwa Racing in 2003 with the intention of setting up a platform where students could gain hands-on experience in vehicle design, manufacturing, and racing technology. He shared an insightful and inspiring talk on the creation and evolution of this team, emphasizing perseverance, teamwork, and innovation. He highlighted the importance of applying theoretical knowledge to real-world design and building critical problem-solving skills.

The seminar concluded with valuable guidance on project planning, technical validation, industry collaboration, and securing sponsorship, motivating students from Ashwa Racing as well as other technical clubs at RV College of Engineering.



Founder Mr. Arunraj Subbaraj addressing the technical clubs of RV College of Engineering.

Bridging Industry and Student Innovation: Workshop Visit by Mr Hubert Reilard

Mr Hubert Reilard, a prominent German-Indian entrepreneur and Managing Partner at COMANO Advisory LLP, shared valuable insights from his three decades of experience and long-standing association with Ashwa Racing.

Mr. Reilard's association with Ashwa Racing dates back to the team's earliest years, reflecting a long-standing commitment to its growth and development. His support can be seen on the team's earliest vehicles, including the RZ-05C, Ashwa Racing's first-ever car, where his company's logo was prominently featured on the car. This partnership continued through later generations of cars such as the RZ-X8C, highlighting the enduring nature of his support. Beyond financial backing, Mr. Reilard has consistently encouraged the team to bridge the gap between theoretical learning and practical engineering.



Mr. Hubert Reilard, with an Ashwa Racing vehicle, X8C model, symbolizing a partnership that has supported and shaped the team across generations.

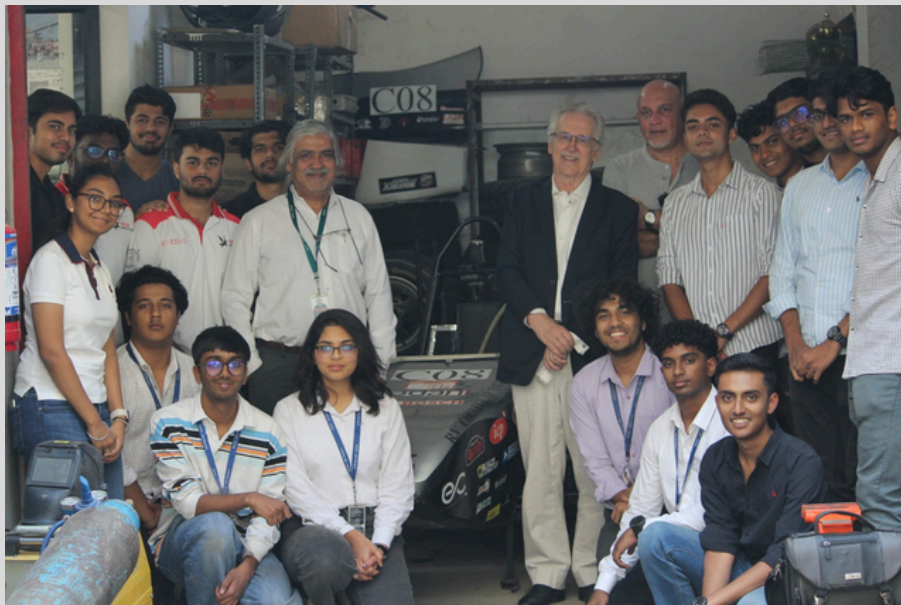


Ashwa Racing's first-ever car, the RZ05 model, reflects the team's foundation years and its early association with EFD.

As an early supporter of Ashwa Racing, Mr Reilard played a pivotal role in the team's growth as one of its first industry investors. His continued support over the years has helped strengthen the team's foundation. He has also facilitated valuable connections with sponsors and partners through his professional network, contributing significantly to the team's development. He echoed the importance of strong mentorship, industry partnerships, and strategic thinking for a successful student innovation platform. We sincerely thank you for your support during the nascent stages of the team.

Engineering Excellence and Validation: Workshop Visit by Harish Kukreja

Mr. Harish Kukreja, a long-standing mentor, visited the workshop to review project progress and provide technical guidance. He focused on validating the vehicle concept to ensure the highest standards of performance, safety, and reliability. Mr. Kukreja offered strategic advice on the technical approach, system integration, rigorous testing, and structured project management for the electric vehicle platform currently under-development. His guidance reinforced the importance of precision and continuous improvement.



Team Ashwa with Mr. Hubert Reilard and Mr. Harish Kukreja.



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**POWERING THE
FUTURE
TOGETHER**

INFINEON
X
ASHWA RACING





INFINEON X ASHWA RACING: POWERING THE FUTURE TOGETHER

Every technological revolution begins with a vision to shape the future. Founded in 1999, Infineon Technologies has grown into one of the world's leading semiconductor companies, shaping what modern electronics is now. From automotive systems and power electronics to embedded security and IoT solutions, Infineon's technologies power many of the devices and systems we use on a daily basis.

Over the years, Infineon has been able to position itself at the centre of the world's move to electrification and digitalisation. This has been achieved by providing semiconductor solutions that make the world a safer place, more efficient, and more connected. Whether it's electric mobility, safety in the automobile industry, or energy efficiency in industrial applications, Infineon has been able to push the limits of technology.

As a team driven by engineering curiosity and the desire to build and think big, we look up to organisations like Infineon that consistently evolve and push the limits of technology. Hence, when we got the opportunity to collaborate with Infineon Technologies as our executive sponsor, we were delighted and honoured.

Our partnership with Infineon Technologies began in 2023, and over the past three years, they have been a cornerstone of our journey. Their unwavering support has played a pivotal role in driving our technical growth and strengthening our capabilities as a team. We extend our heartfelt gratitude to Infineon for their trust and support. Infineon Technologies has been a cornerstone of our journey, playing a pivotal role in our growth and technical development. Their continued support for our electric vehicle initiatives reflects a shared commitment to sustainability and the future of mobility. Together, we aim to accelerate innovation, strengthen our technical capabilities, and build a strong partnership that allows us as a team to grow faster than ever.



अश्व RACING





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**PRECISION
BEHIND OUR
PERFORMANCE**

SANSERA
ENGINEERING
X
ASHWA RACING





SANSERA X ASHWA RACING: PRECISION BEHIND OUR PERFORMANCE

Established in 1981 in Bengaluru, Sansera Engineering began as a dedicated manufacturing organization, which over the years, transformed into one of the leading engineering-driven manufacturers of precision automotive parts in India.

Throughout the years, Sansera has developed itself into an established partner for some of the largest automotive manufacturers across the globe. As an organization that specializes in the manufacturing of high-precision forged and machined parts, they have an important role to play in powering the mobility systems of the modern world. Their parts power the engine, transmissions, and other critical systems that require the highest standards of dependability.

As a student-led mobility team striving to push the boundaries of design and performance, we draw inspiration from organizations that have built their legacy through precision, innovation, and technical excellence.

We would like to thank Sansera Engineering for their trust and confidence in our team. With their support, we look forward to enhancing our engineering capabilities and continuing our journey towards innovation, excellence, and mobility.



SANSERA
ideas@work

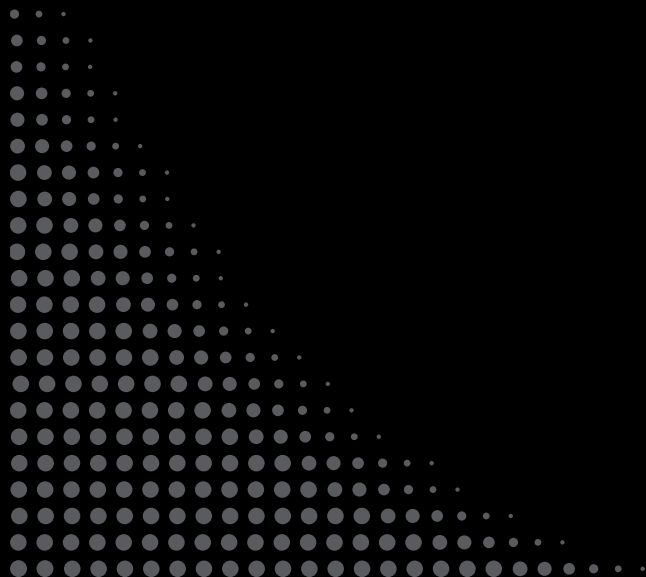


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THE ASSEMBLY PROGRESS

FROM COMPONENTS TO COMPLETE SYSTEM





THE ASSEMBLY PROGRESS

The Electrical Systems team at Ashwa Racing has continued to make steady progress on the integration of both the High Voltage (HV) and Low Voltage (LV) systems on our vehicle.

The work carried out this month primarily involved the completion of the temperature sensing and the cell tap harness routing within the segments in the HV sector. Mounting of the Low Voltage enclosure, harness and routing of the shutdown circuit and CAN communication lines was the progress along the LV sector. Each of these tasks contributes to the safe monitoring, control, and communication of critical electrical systems in the vehicle.

THE HIGH VOLTAGE SYSTEM

The AMS (Accumulator Management System) registers temperature and battery voltage of the modules. Temperature monitoring plays a critical role in the safe operation of our 25R Samsung lithium-ion cells, which are the source of energy in the Accumulator. In case the operation temperature exceeds a given threshold, the car is immediately shut down and the AMS indicator lights up to represent the fault. This is necessary as cells experience significant electrical and thermal loads during operation.

Each Energus module contains temperature sensors that continuously measure the thermal conditions of the cells. These sensors help identify the maximum temperature within each module. During the wiring process, modules are located at varying distances from the segment's central output location, wire lengths were individually measured and prepared accordingly. The harness routing was designed so that all sensor outputs were directed toward the AMS slave board, which collects temperature data and communicates it to the AMS master. The AMS master continuously evaluates this data and ensures that the accumulator operates within safe thermal limits.

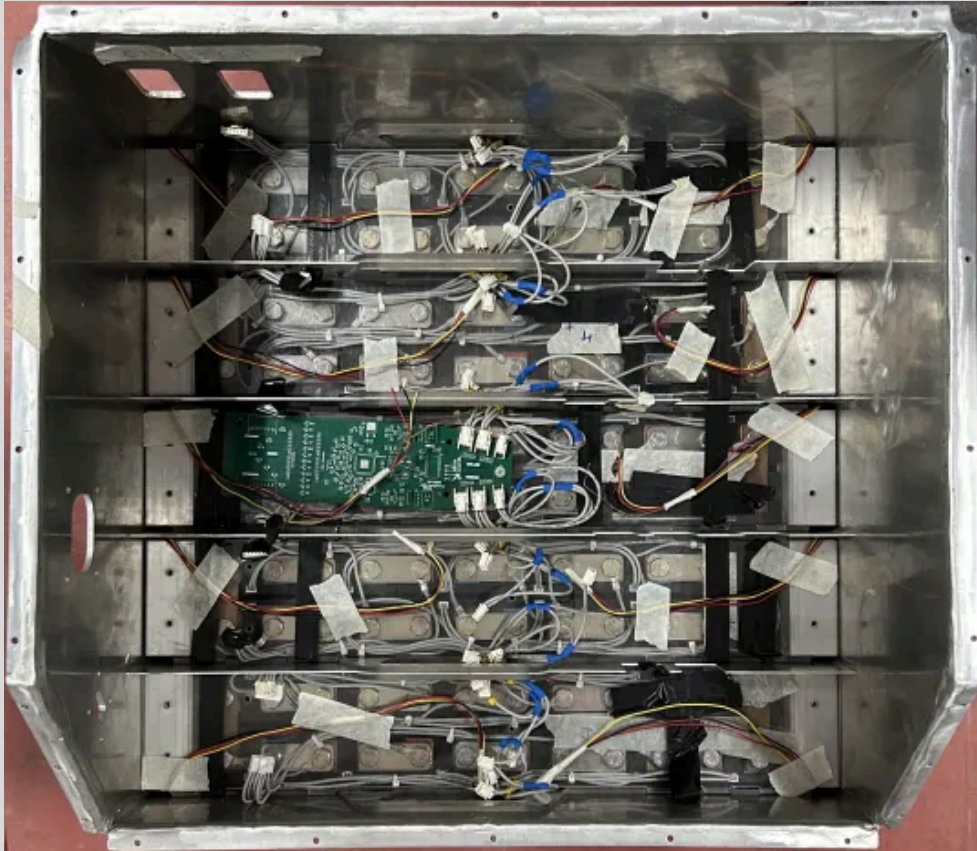


Figure 1- Accumulator with Segments placed within.

The AMS Slave board on top of the middle segment allows for connection of the harness through connectors. Remaining AMS Slaves will be connected in the same fashion.

The wires were routed such that they remained flush with the surface of the segment, ensuring that they do not exceed the height of any bus bars present in the assembly. This prevents interference with other components and ensures that no strain is applied at the end points of the wires. Alongside the temperature harness, the cell tap connections, previously implemented for voltage monitoring, were integrated using similar routing principles.

THE LOW VOLTAGE SYSTEM

On the other hand, progress was also made on the Low Voltage system, beginning with the mounting of the LV enclosure. It was mounted accounting for the grounding. In this arrangement, the ground of the Power Distribution Board (PDB) is connected directly to the chassis through the bolt it is affixed to. This establishes the chassis as the common electrical reference point for the LV circuitry. Proper grounding is essential for maintaining electrical stability, minimizing noise in communication signals, and ensuring consistent operation of sensors and control electronics.

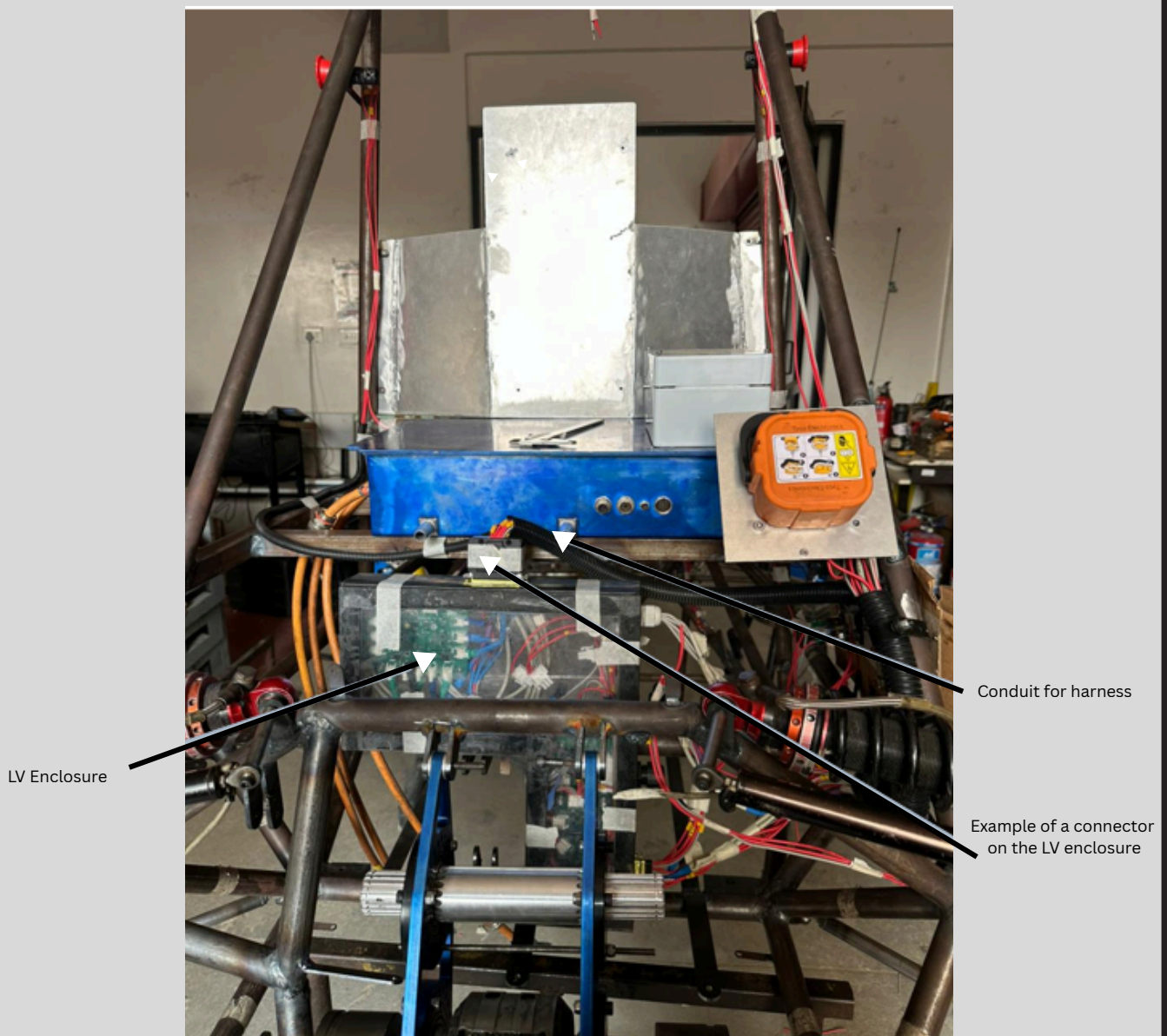


Figure 2- The figure depicts the LV enclosure being mounted on the chassis and shows the wiring harness combined into a conduit.

The shutdown circuit is a critical safety feature designed to immediately cut off power to the motor in the event of a fault. Its reliability is essential for ensuring safe vehicle operation during testing and competition.

To implement this, the required wire lengths between components were measured based on the finalized electrical layout of the vehicle. The routing was performed while accounting for the presence of a firewall separating the driver compartment from the accumulator enclosure, which acts as a protective barrier between the high-energy system and the driver. Where wires needed to pass through the firewall, grommets were used to ensure safe routing. A grommet is a protective ring, typically made of rubber or insulating material, that is inserted into openings where wires pass through structural panels. This design ensures ease of removing the accumulator from the vehicle without having concerns about the wiring. The grommet enables the harness to be divided into two parts- from the cockpit to the firewall and from the firewall to the HV/LV enclosures. Once they are disconnected, the accumulator can be separated from the rest of the vehicle.

The routing of all the wires follow the same pattern - They start at a component 'X', move through the shortest path along the chassis member that does not add extra strain to it, and end at component 'Y'. Once the routing of the wiring is fully completed, the cables are bundled together and enclosed within a protective conduit. The conduits are affixed to the chassis member and form one end of the firewall. Bundling the wires in this manner enhances both safety and maintainability within the vehicle's electrical architecture

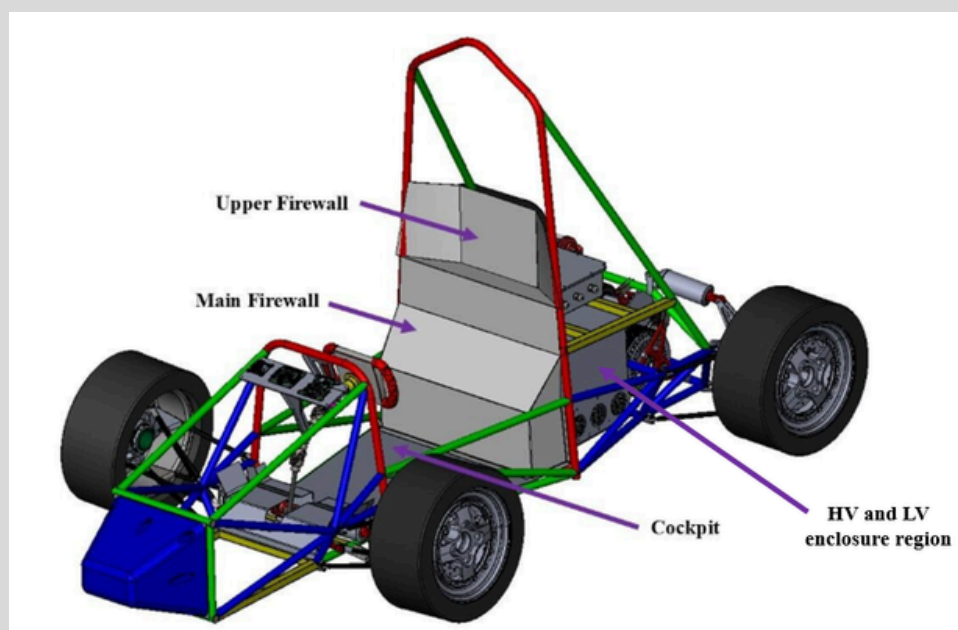


Figure 3 - Location of Firewall and placement of Accumulator behind it.

Harness of CAN (Controller Area Network) wires also took place. CAN (Controller Area Network) is a communication protocol that enables multiple nodes to exchange information over a shared network. The CAN wiring was initiated using dual-core cables provided by LAPP Cables, one of the team's sponsors. The cable structure supports termination impedance characteristics required for reliable data transmission.

Throughout the wiring process, ferrules were used at wire terminations to ensure proper numbering and identification of wires. In order to maintain uniformity, connectors and crimps were standardized across all electrical systems. Adopting such methods proved useful for any member to identify faults and repair them without requiring the presence of other members. These practices also improve organization within the car and have thus simplified troubleshooting, maintenance, and future integration work. As the team transitions their designs into reality, the systems of safety and communication are kept alive throughout the processes.



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(2025-2026)**

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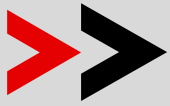
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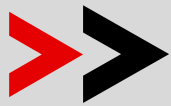


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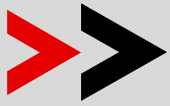
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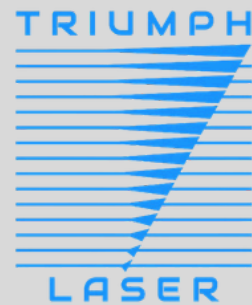
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POWERHAUS



PC PROCESS



DS SOLIDWORKS



TEAM88
INDIA





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DIRECTORY
2026**

DRIVEN BY PASSION, UNITED BY ENGINEERING—
THIS IS ASHWA RACING!

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